Bemnet G. Nega

Atlanta, GA | 404-287-8511 | BemnetgNega@gmail.com | www.linkedin.com/in/bemnetn | US Citizen

Summary

Electrical Engineering student with hands-on experience in electric vehicle high-voltage systems, Li-ion battery integration, embedded electronics, and custom PCB design, using test data to troubleshoot problems and improve designs. Contributed to high-voltage junction box design on Georgia Tech's EcoCAR team and built and tested a battery-powered rover drive system. Led teams as Math Club President and SGA Finance Director. Seeking a Summer 2027 Electrical Engineering internship.

Education

Georgia Institute of Technology | Atlanta, GA Expected May 2028

Bachelor of Science in Electrical Engineering, GPA 3.50/4.00

- *Relevant Coursework:* Circuit Analysis, Digital Logic, Programming for HW/SW Systems, Engineering Computing (MATLAB)

Georgia State University | Atlanta, GA Aug 2024 – May 2026

Transferred with 96 credit hours, GPA 3.70/4.00; Dean's List (Fall 2024, Spring 2025, Spring 2026)

- *Honors:* NASA NCAS Program selectee, Peach State LSAMP Scholar, AMATYC Math Competition winner

Skills

Test & Validation: test setup, data collection, bench testing, design validation, root-cause analysis, oscilloscope
Programming & Data Analysis: Python, MATLAB (data processing and analysis), C/C++ (Arduino), RISC-V
EV & Power: high-voltage junction box design, HV cable routing and grounding, Li-ion battery systems, DC motor drivers, PWM control
Embedded & CAD: ESP32, Raspberry Pi, Altium Designer, KiCad, Autodesk Fusion and Inventor, Onshape, NI Multisim
Languages: English (fluent), Amharic (native)

Experience

STORCAPP, Washington, D.C. Aug 2025 – Aug 2026

Embedded Systems Engineering Intern

- Designed, prototyped, and tested embedded electronics (hardware and firmware) for a maternal health device, progressing from bench testing to a custom PCB designed in Altium Designer.
- Integrated power systems and sensors, resolving 4 hardware and firmware issues through bench testing and root-cause analysis, improving reliability and safety.

CATLAB, Atlanta, GA Aug 2024 – Present

Makerspace Lab Technician

- Lead workshops on soldering, 3D printing, and CAD, communicating technical processes clearly to students and faculty and training them on engineering software, safe use of fabrication equipment, and rapid prototyping workflows.
- Troubleshoot electronics, 3D printers, computers, and software, and maintain lab equipment to keep the lab ready for prototype fabrication and testing.

Projects

EcoCAR VIP | *Georgia Tech*

- Member of a collegiate team designing and validating an electrified vehicle.
- Contributed to high-voltage junction box design, evaluating placement against HV cable routing, grounding, and competition electrical safety rules.

Autonomous Soil-Sampling Rover | *CORE Program* Summer 2026

- Designed and assembled the drive system on a custom aluminum and 3D-printed chassis, integrating six DC motors, four steering servos, and RC wireless control.
- Integrated dual 11.1V Li-ion battery packs with a custom PCB for power distribution and motor control (PWM via PCA9685).
- Validated mobility, steering performance, and mechanical reliability by collecting and analyzing test data, using the results to guide design changes.

Leadership & Activities

Math Club | President, Georgia State University Jan 2026 – May 2026

- Led team to 1st place (2025) and 2nd place (2026) at the University of North Georgia Math Competition.

Student Government Association | Finance Director, Georgia State University Aug 2025 – Feb 2026

- Managed a $21,000 budget and reviewed funding requests for student organizations.

National Society of Black Engineers (NSBE) | Member